

SUHAIL MAHMUD

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PROFESSIONAL SUMMARY

Computational scientist and ML practitioner with 10+ years working at the intersection of atmospheric modeling, climate data, and machine learning. PhD research centered on boundary layer physics and WRF-based numerical weather prediction, with applied ML methods for atmospheric process emulation. Subsequent industry work spans probabilistic catastrophe modeling for natural hazards including tropical cyclones, winter storms, flood, and wildfire. Proficient in Python, PyTorch, xarray, NetCDF, and geospatial data processing. Experienced deploying ML workflows on HPC systems and communicating model results to non-technical audiences.

TECHNICAL SKILLS

Programming: Python (PyTorch, scikit-learn, NumPy, pandas), R, Fortran (WRF)

Climate & Geospatial Data: xarray, NetCDF, GRIB, reanalysis (ERA5, NCEP), NWP outputs, satellite data, WRF model output

Geospatial Tools: QGIS, Shapely, GeoPandas, rasterio, Cartopy, GIS raster/vector workflows

ML Methods: Spatio-temporal ML, regression (Ridge, Lasso, GLM), hurdle models, convolutional and recurrent neural networks, random forests, XGBoost, ensemble methods

HPC & Infrastructure: XSEDE/HPC job scheduling (SLURM), MPI parallelism, large-ensemble simulation, AWS (S3, EC2), Git, bash scripting

Data Engineering: ETL pipeline design, Snowflake, Databricks, dbt, PostgreSQL, data quality control

Visualization: matplotlib, plotly, Tableau, Power BI

PROFESSIONAL EXPERIENCE

Senior Data Scientist | [KatRisk LLC](#)

Aug 2024 – Jan 2026

- Built probabilistic hazard and vulnerability models for wildfire, flood, and storm surge perils, contributing to stochastic event catalog workflows used by reinsurance clients.
- Developed ML-based damage ratio estimation models by combining geospatial exposure layers with historical loss data, improving loss curve accuracy for portfolio risk assessment.
- Processed and quality-controlled large geospatial hazard datasets (raster grids, binary LZ4-compressed files) for use in downstream probabilistic modeling and cat model scoring.
- Collaborated with meteorologists and hydrologists to integrate physical hazard insights into data-driven modeling frameworks, ensuring scientific consistency in peril representations.
- Maintained version-controlled model codebases using Git and documented workflows for reproducibility across research and production environments.
- Prepared and presented model methodology and validation results to non-technical stakeholders, translating probabilistic outputs into actionable risk summaries.

Data Engineer | [Tomorrow.io](#)

Aug 2023 – Aug 2024

- Engineered cloud-based pipelines on AWS to ingest and process high-resolution NWP forecast outputs and observational weather data for real-time risk analytics products.
- Built ETL workflows to clean, transform, and load multi-source atmospheric and satellite datasets into Snowflake, supporting downstream ML feature pipelines.
- Developed data quality control checks for incoming weather data streams, identifying and handling gaps, outliers, and format inconsistencies across global observation networks.
- Wrote modular, well-documented Python code across version-controlled repositories, with peer code review as a standard part of the development process.

Senior Research Scientist | [Karen Clark & Company](#)

May 2022 – Aug 2023

- Contributed to stochastic event catalog development for tropical cyclone, winter storm, and severe convective storm perils, applying statistical and physical modeling methods.

- Led historical loss analysis and vulnerability model validation, benchmarking results against third-party cat models and industry datasets to assess model performance on real-world data.
- Applied regression and clustering methods to improve damage function estimation across diverse building stock and exposure portfolios.
- Presented technical findings and model validation results to clients and leadership, including audiences without quantitative modeling backgrounds.

Postdoctoral Scholar | [Penn State University](#)

Jan 2021 – Apr 2022

- Continued research on atmospheric boundary layer dynamics using WRF large-eddy simulation, focusing on turbulence structure, surface flux parameterization, and model validation against observational data.

Research Assistant (PhD) | [University of Texas at El Paso](#)

Aug 2014 – Dec 2020

- Developed ML surrogate models to emulate atmospheric boundary layer processes simulated by WRF, using supervised learning to accelerate large-ensemble runs on XSEDE HPC systems.
- Processed and analyzed multi-dimensional climate and atmospheric datasets in NetCDF and HDF formats using Python, applying feature engineering and dimensionality reduction for spatio-temporal modeling.
- Built data pipelines for ingesting reanalysis and NWP outputs, with quality control and preprocessing steps for wind, temperature, and turbulence variables across vertical atmospheric profiles.
- Collaborated with a multidisciplinary research team including atmospheric scientists and engineers, contributing to model development and joint publications.
- Authored 3 peer-reviewed publications on ML-based atmospheric boundary layer modeling; presented at SIAM and AMS national conferences.

EDUCATION

Ph.D., Computational Science *2020*

University of Texas at El Paso | Dissertation: Atmospheric Boundary Layer Modeling using ML and WRF

M.S., Computational Science *2016*

University of Texas at El Paso

PUBLICATIONS & AFFILIATIONS

- 3 peer-reviewed publications on ML-based atmospheric boundary layer modeling using WRF and numerical weather prediction data.
- XSEDE/HPC training certifications in large-scale scientific computing and parallel workflows.
- Member, Society for Industrial and Applied Mathematics (SIAM); Member, American Meteorological Society (AMS).